

Task 2 - API Integration

Weather Information

Objective : A simple Python application that fetches real-time weather information for any city using the OpenWeatherMap API. The application allows users to search for a city and displays current weather details such as temperature, humidity, wind speed, and weather conditions..

Technologies Used: Python 3, Requests Library ,OpenWeatherMap API

Features : Search weather by city name, Fetch live weather data from an API, Parse JSON responses, Display temperature, humidity, pressure, and wind speed, Exception handling for invalid cities and network errors, Easy-to-use command-line interface.

Code Snippets

```
requirements.txt weather.py X
weather.py > get_weather
20
21 def get_weather(city):
22     """Fetch and display weather information for a city."""
23
24     url = f"{BASE_URL}?q={city}&appid={API_KEY}&units=metric"
25
26     try:
27         response = requests.get(url, timeout=10)
28
29         if response.status_code == 200:
30             data = response.json()
31
32             current_time = datetime.now().strftime("%d-%m-%Y %H:%M:%S")
33
34             print("\n===== Weather Report =====")
35             print(f>Date & Time : {current_time}")
36             print(f>City : {data['name']}")
37             print(f>Country : {data['sys']['country']}")
38             print(f>Temperature : {data['main']['temp']} °C")
```

After running the code :

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
PS C:\Users\dinod\Desktop\Python Projects\Task-2-API-Integration-Weather> python weather.py
Enter city name (or 'exit'): Indore

===== Weather Report =====
Date & Time : 12-07-2026 11:45:54
City       : Indore
Country    : IN
Temperature : 27.34 °C
Feels Like : 28.87 °C
Humidity    : 64 %
Pressure    : 1004 hPa
Weather     : Overcast Clouds
Wind Speed  : 7.26 m/s
Visibility  : 10.0 km
=====

Enter city name (or 'exit'): exit
Goodbye!
❖ PS C:\Users\dinod\Desktop\Python Projects\Task-2-API-Integration-Weather> 
```

Repository Link :

<https://github.com/ChetanSonare666/Task-2-API-Integration-Weather>